

Regular Perturbation and an Exact ODE Solution

Methods of Applied Mathematics

Homework 1, Problem 2

Selected solution, edited for standalone reading

Consider the parameter-dependent initial-value problem

$$y'' + 2\varepsilon y' - y = 0, \quad y(0) = 1, \quad y'(0) = 0. \quad (1)$$

The exact solution provides a check on the regular perturbation expansion $y = y_0 + \varepsilon y_1 + \dots$ for small ε . Both routes yield the same first-order correction.

Exact solution

We try the solution in the form

$$y(t) = A_1 e^{\lambda_1 t} + A_2 e^{\lambda_2 t}.$$

Substituting an exponential into the differential equation gives the characteristic polynomial. Hence $\lambda_{1,2}$ are roots of the quadratic equation

$$x^2 + 2\varepsilon x - 1 = 0,$$

so that

$$\lambda_{1,2} = -\varepsilon \pm \sqrt{\varepsilon^2 + 1}.$$

Now impose the initial conditions:

$$y(0) = A_1 + A_2 = 1, \quad y'(0) = A_1 \lambda_1 + A_2 \lambda_2 = 0.$$

Solving for A_1, A_2 , we obtain

$$A_1 = \frac{\varepsilon^2 + 1 + \varepsilon\sqrt{\varepsilon^2 + 1}}{2(\varepsilon^2 + 1)}, \quad A_2 = \frac{\varepsilon^2 + 1 - \varepsilon\sqrt{\varepsilon^2 + 1}}{2(\varepsilon^2 + 1)}.$$

Therefore, with $s = \sqrt{1 + \varepsilon^2}$, the exact solution can be written compactly as

$$y(t) = e^{-\varepsilon t} \left[\cosh(st) + \frac{\varepsilon}{s} \sinh(st) \right].$$

First-order regular perturbation

Substitute the expansion

$$y(t) \sim y_0(t) + \varepsilon y_1(t) + \dots$$

into the ODE (1). We obtain

$$(y_0'' - y_0) + \varepsilon (y_1'' + 2y_0' - y_1) + \varepsilon^2 (2y_1' + \dots) + \dots = 0.$$

Equating coefficients order by order:

Order $O(1)$:

$$y_0'' - y_0 = 0, \quad y_0(0) = 1, \quad y_0'(0) = 0.$$

Solution:

$$y_0(t) = \cosh(t).$$

Order $O(\varepsilon)$:

$$y_1'' - y_1 = -2y_0', \quad y_1(0) = 0, \quad y_1'(0) = 0.$$

Since $y_0'(t) = \sinh(t)$, this becomes

$$y_1'' - y_1 = -2\sinh(t).$$

A particular solution is $y_{1p}(t) = -t \cosh(t)$. The homogeneous solution is $C_1 \cosh(t) + C_2 \sinh(t)$. So

$$y_1(t) = C_1 \cosh(t) + C_2 \sinh(t) - t \cosh(t).$$

Applying $y_1(0) = 0 \implies C_1 = 0$, and $y_1'(0) = 0 \implies C_2 = 1$, we get

$$y_1(t) = \sinh(t) - t \cosh(t).$$

Thus, up to $O(\varepsilon)$,

$$\boxed{y(t) \sim \cosh(t) + \varepsilon(\sinh(t) - t \cosh(t))}.$$

Comparison with the exact solution

For small ε , the coefficients and exponents satisfy

$$A_1 = \frac{1}{2} + \frac{1}{2}\varepsilon + O(\varepsilon^2), \quad A_2 = \frac{1}{2} - \frac{1}{2}\varepsilon + O(\varepsilon^2),$$

$$\lambda_1 = 1 - \varepsilon + O(\varepsilon^2), \quad \lambda_2 = -1 - \varepsilon + O(\varepsilon^2).$$

For fixed t , or uniformly on a bounded t -interval, substituting these expansions gives the same first-order approximation. Thus

$$\boxed{y(t) = \cosh t + \varepsilon(\sinh t - t \cosh t) + O(\varepsilon^2)}.$$

This agrees with the first-order regular perturbation calculation.